

# estim8 - An FMI-compliant Python toolbox for bioprocess modeling and parameter estimation

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## Software

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## Summary

Modeling and simulation are indispensable tools for understanding the complex nature of biological systems and making the most of the information contained in experimental data. Functional integration of quantitative measurement data and rigorous parameter estimation are essential for creating valid, predictive models. We present `estim8`, a Python package for FMI-compliant ODE and DAE (bio)process modeling and parameter estimation. It offers convenient handling of multiple experimental replicates coupled with highly scalable solutions using federated simulation operations. The source code is available on [GitHub](#). The package is unit-tested on Windows and Linux. Thorough documentation is available under <https://estim8.readthedocs.io> including various example notebooks.

## Statement of Need

Mathematical modeling has become a pivotal tool in biotechnological research and industrial bioprocess development, supporting the analysis and interpretation of complex experimental data (Fischer, 2008; Hartmann et al., 2022; Noll & Henkel, 2020; Ploch et al., 2019). While ordinary differential equations (ODEs) are commonly used to describe continuous biological systems, many biotechnological applications require differential algebraic equation (DAE) systems to handle discontinuities, discrete events, physical constraints, and embedded optimization criteria (Ploch et al., 2020).

A crucial step in the modeling workflow is parameter estimation – or in layman’s terms “fitting the model”. This step questions the theoretical understanding of the system under investigation using real data, ultimately leading to confirmation or falsification of the hypotheses put forward. Although several general-purpose software tools for model formulation, simulation and parameter estimation exist, they currently present various limitations for biotechnological applications, particularly regarding DAE support, handling of experimental replicates, and accessibility.

To address these limitations, we present `estim8`: a Python-based toolbox for simulation and parameter estimation of dynamic models. It is built on the Functional Mock-up Interface (FMI) standard (Modelica Association, 2023) and employs metaheuristic algorithms for optimization problems. `estim8` provides specialized functionality for biotechnological applications, particularly in handling experimental replicates. By supporting model definition and simulation export from various FMI-compliant third-party software, including the open source OpenModelica platform (Fritzson et al., 2020), `estim8` enables comprehensive DAE support and convenient event handling. `estim8` is designed to provide straightforward access to modeling workflows

41 for domain experts in biotechnological research without requiring extensive computational  
42 expertise.

## 43 State of the Field

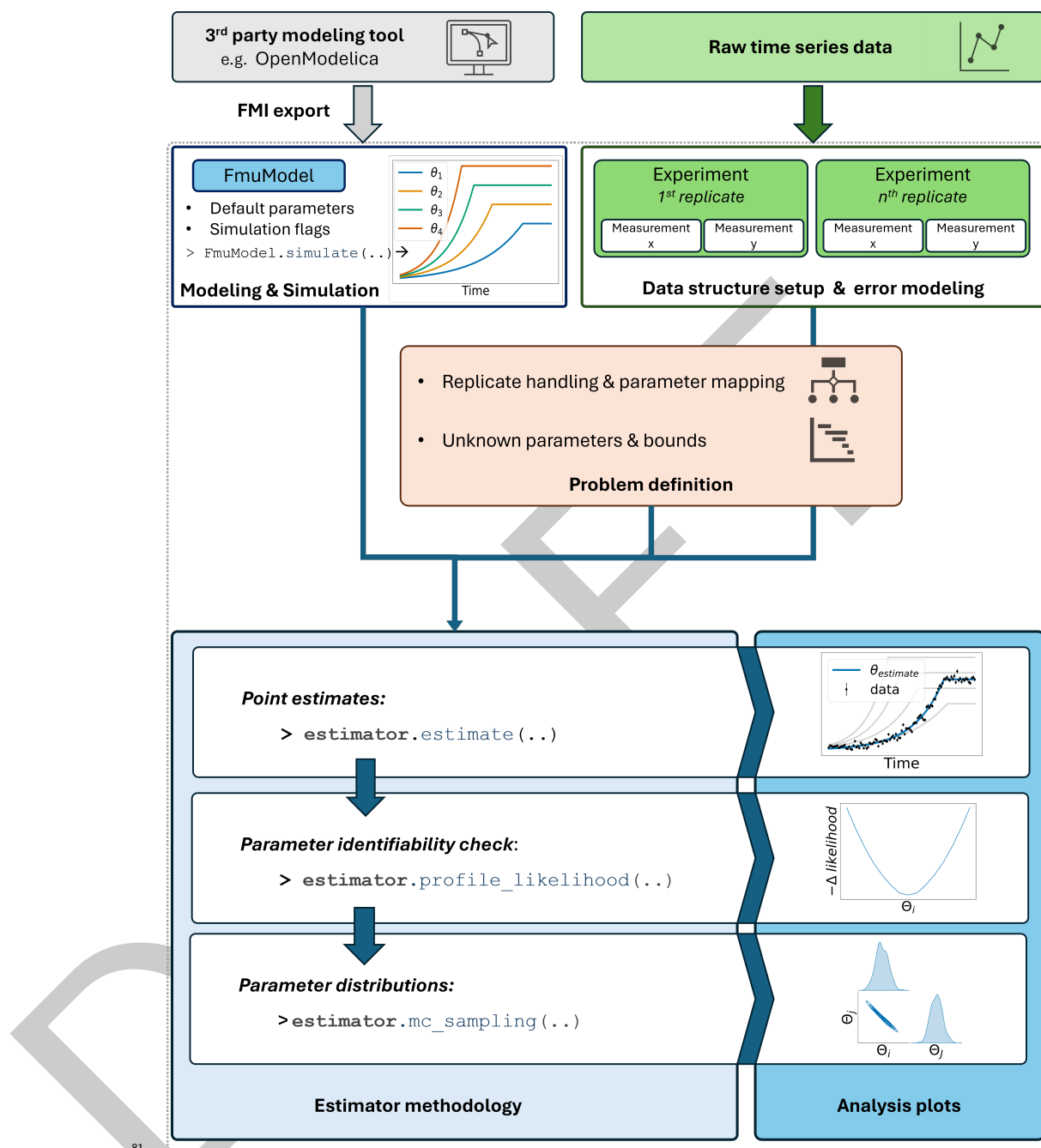
44 Many established tools such as pyFOOMB (Hemmerich et al., 2021), COPASI (Hoops et al.,  
45 2006), PyBioNetFit (Mitra et al., 2019), and Data2Dynamics (Raue et al., 2015) are limited  
46 to ODE systems, leaving users without native support for the DAE formulations often required  
47 in biotechnological process modeling. DAE Tools (Nikolić, 2016) and ModestPy (K. Arendt,  
48 2018) support DAE systems but lack dedicated functionality for common biotechnological  
49 workflows such as structured handling of experimental replicates across different conditions,  
50 and therefore require substantial workarounds. The combination of AMICI (Fröhlich et al.,  
51 2021) and pyPESTO (Schälte et al., 2023) offers high-performance sensitivity analysis, but  
52 poses a relatively high entry barrier with respect to model formulation, making it less accessible  
53 to domain experts without extensive computational background. estim8 combines FMI-based  
54 DAE support, dedicated replicate handling, and an accessible Python interface to address these  
55 gaps.

## 56 Workflow

57 The workflow in estim8 follows a structured approach to bioprocess modeling and parameter  
58 estimation (Figure 1). Users start by developing mathematical models using third-party  
59 software that supports the FMI standard, such as OpenModelica (Fritzson et al., 2020). This  
60 open-source platform provides an interactive modeling environment with graphical features  
61 and allows models to be exported as Functional Mock-up Units (FMUs), supporting both  
62 CoSimulation and ModelExchange formats. Notably, models of the SBML (Hucka et al., 2003)  
63 standard can be translated to Modelica (Maggioli et al., 2019), further expanding the range of  
64 modeling tools supported by estim8.

65 The exported FMU is then loaded into estim8 using the FmuModel class. The package  
66 implements a structured data hierarchy where Experiment objects contain Measurement objects  
67 with an associated error\_model and an observation\_mapping. A key feature of estim8  
68 is its comprehensive handling of biological replicates, which are crucial for the statistical  
69 quality of data from biological experiments (Casler et al., 2015). Based on a user-defined  
70 ParameterMapping, common properties of replicates and different conditions between them can  
71 be modeled by defining so-called global and local parameters. This concept utilizes redundant  
72 information in the measurement data, thereby effectively reducing the number of parameters  
73 to be estimated (Helleckes et al., 2022; Hemmerich et al., 2021; Osthege & Helleckes, 2022).

74 The core of estim8 is the Estimator class, which serves as a central hub for managing parameter  
75 estimation tasks. This class stores all data entered by the user and provides functions for solving  
76 optimization problems. In addition, identifiability analyses and uncertainty quantifications can  
77 be performed using profile likelihood or Monte Carlo sampling. The visualization module  
78 offers comprehensive visualization methods for analyzing simulation results, comparing model  
79 predictions with experimental data, and evaluating parameter estimation results. A thorough  
80 guideline is given by several example notebooks in our [documentation](#).



**Figure 1:** Schematic overview of the estim8 workflow. A dynamic model is developed in a third-party FMI-compliant modeling tool (e.g. OpenModelica) and loaded into estim8 as an FmuModel. Experimental time series data from possibly multiple replicates is organized into Experiment and Measurement objects. The estimation problem is defined by specifying a ParameterMapping — distinguishing global parameters shared across replicates from local replicate-specific parameters — along with parameter bounds. The Estimator class provides three levels of analysis: point estimates via metaheuristic optimization (estimate()), parameter identifiability assessment via profile likelihood (profile\_likelihood()), and full parameter distributions via Monte Carlo sampling (mc\_sampling()), each with corresponding analysis plots.

## 92 Software Design

93 estim8 is an open-source Python package compatible and tested with Windows and Linux/Unix  
 94 platforms. The modular, object-oriented architecture allows for easy expansion by new  
 95 implementations, e.g. custom simulators or cost functions. At the very core, estim8 currently  
 96 features interactive simulation of Functional Mock-up Units (FMUs) via the ModelExchange  
 97 or CoSimulation interface in Python utilizing FMPy (Sommer, 2020). The embedded SciPy  
 98 (Virtanen et al., 2020) and pygmo (Izzo & Dario, 2020) packages offer a variety of optimization  
 99 algorithms that can be used for parameter estimation. Additionally, estim8 supports error  
 100 propagation in measurement data and quantification of uncertainties in model predictions.

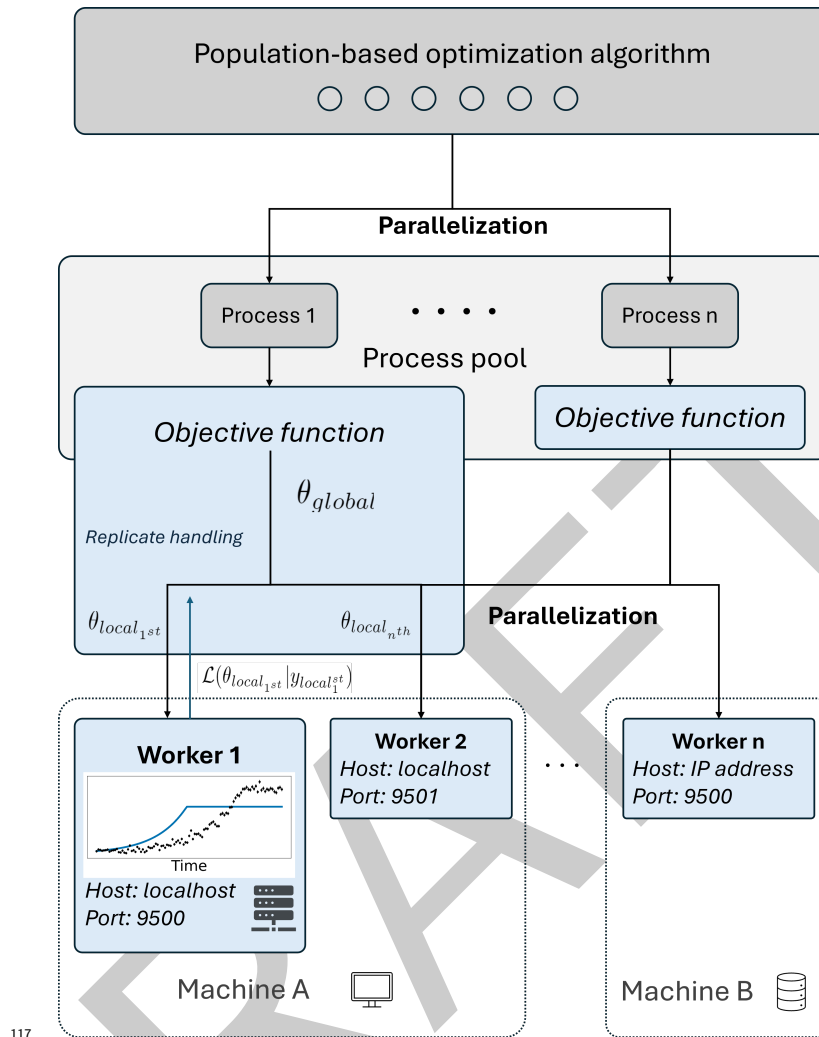
## 101 Scalability

102 Parameter estimation requires numerous simulation steps for evaluating a candidate solution  
 103  $\theta$  by comparing the resulting model predictions to experimental data based on a statistical  
 104 likelihood measure  $\mathcal{L}$ . This procedure – commonly referred to as *evaluation of the objective*  
 105 *function* – is the time-determining step in most applications. Many population-based solvers  
 106 therefore enable parallel evaluations of the objective function; the pygmo package in particular  
 107 allows for highly parallelizable setups (grey parts of Figure 2).

108 However, the integration of experimental replicates significantly increases the number of  
 109 simulations, as the objective function for a global parameter set  $\theta_{global}$  is now a differentiable  
 110 function resulting from the sum of replicate-specific (local) likelihood measures:

$$\mathcal{L}(\theta_{global}|y_{global}) = \sum_{i=1}^n \mathcal{L}(\theta_{local_i}|y_{local_i})$$

111 To this end, estim8 provides the option to use a so-called federated computing setup (Figure  
 112 2), which effectively introduces an additional parallelization layer. Using pytensor-federated  
 113 (Ostehege, 2023), the computation of differentiable objective functions is distributed via gRPC  
 114 streams to federated worker nodes which carry out the simulation tasks. The worker nodes  
 115 can therefore be launched on different machines in a computer cluster. This allows for massive  
 116 parallelization of computationally expensive model units.



**Figure 2:** Federated computation setup for differentiable objective functions. A population-based optimization algorithm distributes candidate global parameter sets  $\theta_{global}$  across a process pool for parallel evaluation. Within each process, the objective function performs replicate handling by dispatching replicate-specific local parameter sets  $\theta_{local}$  to federated worker nodes via gRPC streams. Each worker independently simulates the FMU and returns its replicate-specific likelihood contribution  $\mathcal{L}(\theta_{local}|y_{local})$ . Workers can be deployed across multiple machines (e.g. in a compute cluster), introducing an additional parallelization layer on top of the process pool for computationally expensive models.

## Limitations

Currently, `estim8` does not incorporate gradient-based optimization algorithms, which could enhance parameter estimation efficiency through parametric sensitivities (Villaverde et al., 2018). This capability could be implemented once OpenModelica supports FMI 3.0 (Modelica Association, 2022), providing access to adjoint derivative functions, which are essential for efficiently computing gradients in high-dimensional parameter spaces. Future developments include the integration of Bayesian optimization methods from packages like PyMC (Abril-Pla et al., 2023) and hopsy (Paul et al., 2024).

## Research impact statement

estim8 introduces a streamlined bioprocess modeling workflow that makes rigorous hypothesis testing accessible to domain experts without extensive computational background. By combining accessible Python interfaces and FMI-compliant modeling software such as OpenModelica, estim8 allows researchers to focus on the scientific question rather than the computational implementation. With comprehensive DAE support and tailored solutions for handling experimental replicates — which are increasingly relevant in the context of laboratory automation — estim8 enables rapid iteration through cycles of model evaluation, fitting, and falsification. This lowers the barrier for model-based reasoning in biotechnological applications, supporting faster translation of experimental observations into quantitative process understanding.

## AI usage disclosure

Claude 3.5 Sonnet was occasionally used during software development for drafting and refining implementation ideas. No AI-generated code was incorporated into the software. All design decisions, code editing and reviews were conducted by the researchers. The authorship of this manuscript was conducted without the use of generative AI tools.

## Author contributions

estim8 was conceptualized by DS, SN and TL. Software developments were conducted by TL and DS, critical review of architecture decisions and contribution to the distributed computing setup by MO. The original draft was written by TL, editing and review was done by SN and MO. The work was supervised by SN and funding was acquired by SN and WW.

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## Competing interests

No competing interest is declared.

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